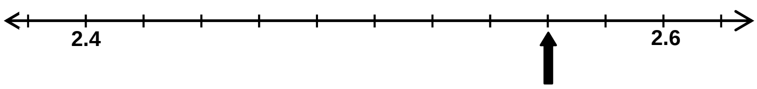
Locating rational and irrational numbers on the number line

Teacher resource

Question 1



The numbers 2.4 and 2.6 are shown on this number line.

Name the number that is indicated by the arrow.

Answer: 2.56
Skill: Students read scales with not all calibrations labelled.

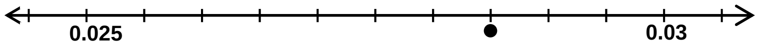
Background information
Students may need to be prompted to draw on their understanding of multiplication and division, rather than rely on trial and error strategies using addition, to calculate the size of the intervals between labelled marks on a scale.

Students should also understand that the multiplicative relationships between the places in the whole number place value system are continued when the whole number place value system is continued to include decimals. This supports students to understand how decimals can be used to represent numbers on a number line between whole numbers and between two decimals numbers on a number line.

Additional questions

1. On a scale, what is the size of each step if there are 5 steps to get from 10 to 20?
2. On a scale, what is the size of each step if there are 10 steps to get from 4 to 5?
3. Each step on a scale is 0.02. How many steps will a scale show between 4.6 and 4.8?

Question 2



The numbers 0.025 and 0.03 are shown on this number line.

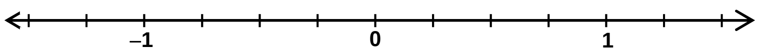
Name the number that is indicated by the point on the line.

Answer: 0.0285
Skill: Students interpret a decimal scale with not all calibrations labelled.

Additional questions

1. If the number 0.025 in the diagram was replaced with -0.02, what number would be indicated by the point?
2. If the number 0.03 in the diagram was replaced with 0.026, what number would be indicated by the point?

Question 3



The numbers -1, 0 and 1 are shown on this number line.

Which number is exactly half-way between $-\frac{1}{2}$ and $\frac{1}{4}$?

$-\frac{1}{3}$	$-\frac{1}{4}$	$-\frac{1}{6}$	$-\frac{1}{8}$
O	O	O	O

Answer: D
Skill: Students calculate the mean of two fractions and locate this on a number line.

Background information
Students may be able to express fractions as equivalent fractions with different denominators but not draw on this understanding to order, compare or find a number between two fractions on a number line.

Additional questions

1. What number is halfway between $1\frac{3}{4}$ and $2\frac{3}{4}$?
2. What number is halfway between $2\frac{3}{4}$ and $4\frac{1}{2}$?
3. The number $2\frac{7}{8}$ is halfway between $1\frac{1}{3}$ and which number?

Question 4

The four numbers π, -π, 7% and 73% are marked on a number line.

Which pair of numbers is furthest apart on the number line?

O 7% and 73%

O π and –π

O -π and 73%

O 7% and π

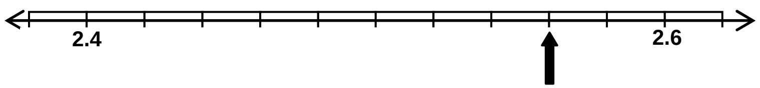
Answer: B
Skill: Students locate positions of rational and irrational numbers on a number line.

Additional questions

1. What is the distance between the points (π+1) and (π-5) on a number line?
2. How do the distances between 0 and π/6 on a number line and 0 and π/2 on a number line compare?

Student activity: Locating rational and irrational numbers on the number line

Question 1



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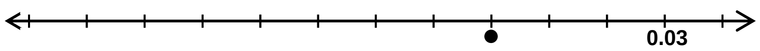

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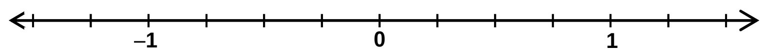

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